

Experiment No.9: Process Explorer

Aim

To use Process Explorer to examine running processes, analyze CPU and memory usage, inspect process properties, check network activity, and identify potentially suspicious processes.

Software Required

- Windows Operating System
- Process Explorer (Sysinternals)
- Windows Security
- Web Browser

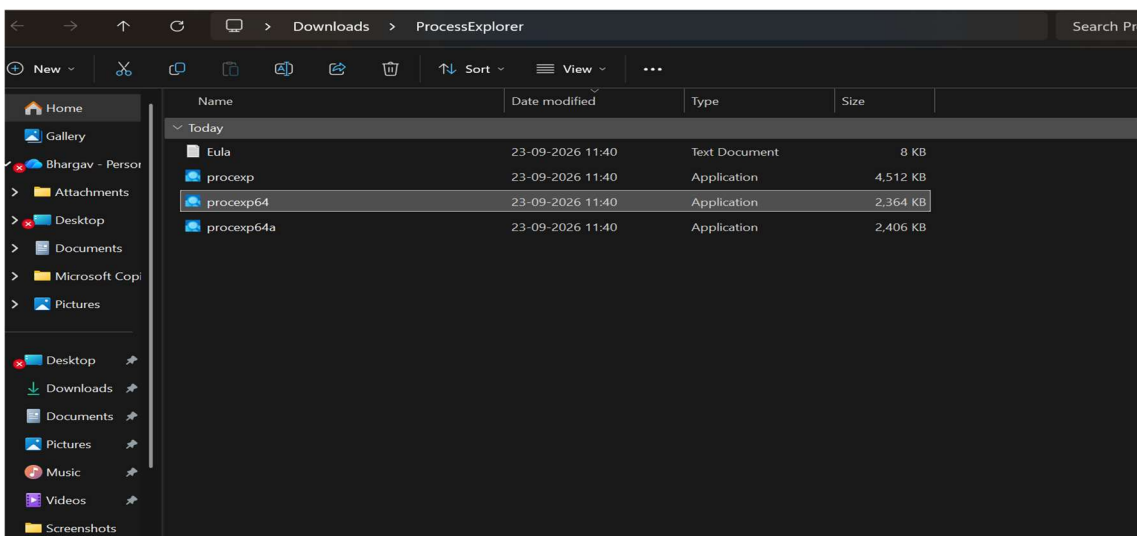
Procedure

Step 1 – Download and Extract Process Explorer

Process Explorer was downloaded and extracted into the 'ProcessExplorer' folder.

The folder contained:

- procexp.exe
- procexp64.exe
- procexp64a.exe
- Eula.txt



Step 2 – Run Process Explorer as Administrator

The 'procxp64.exe' application was executed using ****Run as administrator****.

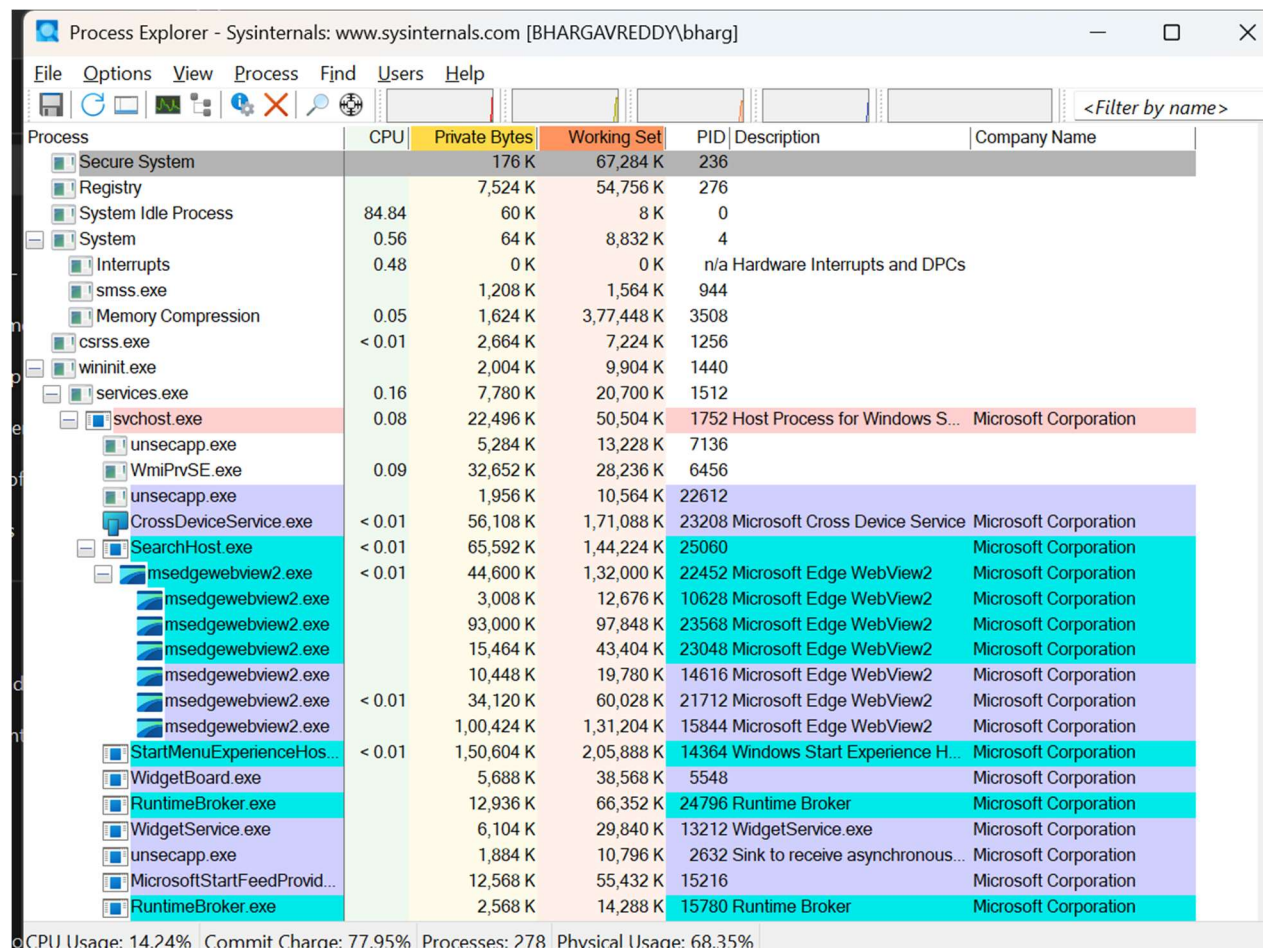
Process Explorer displayed the running processes in a hierarchical process tree.

Step 3 – Examine Running Processes

The Process Explorer window was examined using the following columns:

- Process
- CPU
- Private Bytes
- Working Set
- PID
- Description
- Company Name

The process tree was inspected to identify processes and their resource usage.



The screenshot shows the Process Explorer window with the following columns: Process, CPU, Private Bytes, Working Set, PID, Description, and Company Name. The process tree on the left shows the hierarchy starting from System, with various system and user processes listed. The table on the right provides detailed resource usage for each process.

Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
Secure System		176 K	67,284 K	236		
Registry		7,524 K	54,756 K	276		
System Idle Process	84.84	60 K	8 K	0		
System	0.56	64 K	8,832 K	4		
Interrupts	0.48	0 K	0 K	n/a	Hardware Interrupts and DPCs	
smss.exe		1,208 K	1,564 K	944		
Memory Compression	0.05	1,624 K	3,77,448 K	3508		
csrss.exe	< 0.01	2,664 K	7,224 K	1256		
wininit.exe		2,004 K	9,904 K	1440		
services.exe	0.16	7,780 K	20,700 K	1512		
svchost.exe	0.08	22,496 K	50,504 K	1752	Host Process for Windows S...	Microsoft Corporation
unsecapp.exe		5,284 K	13,228 K	7136		
WmiPrvSE.exe	0.09	32,652 K	28,236 K	6456		
unsecapp.exe		1,956 K	10,564 K	22612		
CrossDeviceService.exe	< 0.01	56,108 K	1,71,088 K	23208	Microsoft Cross Device Service	Microsoft Corporation
SearchHost.exe	< 0.01	65,592 K	1,44,224 K	25060		Microsoft Corporation
msedgeview2.exe	< 0.01	44,600 K	1,32,000 K	22452	Microsoft Edge WebView2	Microsoft Corporation
msedgeview2.exe		3,008 K	12,676 K	10628	Microsoft Edge WebView2	Microsoft Corporation
msedgeview2.exe		93,000 K	97,848 K	23568	Microsoft Edge WebView2	Microsoft Corporation
msedgeview2.exe		15,464 K	43,404 K	23048	Microsoft Edge WebView2	Microsoft Corporation
msedgeview2.exe		10,448 K	19,780 K	14616	Microsoft Edge WebView2	Microsoft Corporation
msedgeview2.exe	< 0.01	34,120 K	60,028 K	21712	Microsoft Edge WebView2	Microsoft Corporation
msedgeview2.exe		1,00,424 K	1,31,204 K	15844	Microsoft Edge WebView2	Microsoft Corporation
StartMenuExperienceHos...	< 0.01	1,50,604 K	2,05,888 K	14364	Windows Start Experience H...	Microsoft Corporation
WidgetBoard.exe		5,688 K	38,568 K	5548		Microsoft Corporation
RuntimeBroker.exe		12,936 K	66,352 K	24796	Runtime Broker	Microsoft Corporation
WidgetService.exe		6,104 K	29,840 K	13212	WidgetService.exe	Microsoft Corporation
unsecapp.exe		1,884 K	10,796 K	2632	Sink to receive asynchronous...	Microsoft Corporation
MicrosoftStartFeedProvid...		12,568 K	55,432 K	15216		Microsoft Corporation
RuntimeBroker.exe		2,568 K	14,288 K	15780	Runtime Broker	Microsoft Corporation

CPU Usage: 14.24% | Commit Charge: 77.95% | Processes: 278 | Physical Usage: 68.35%

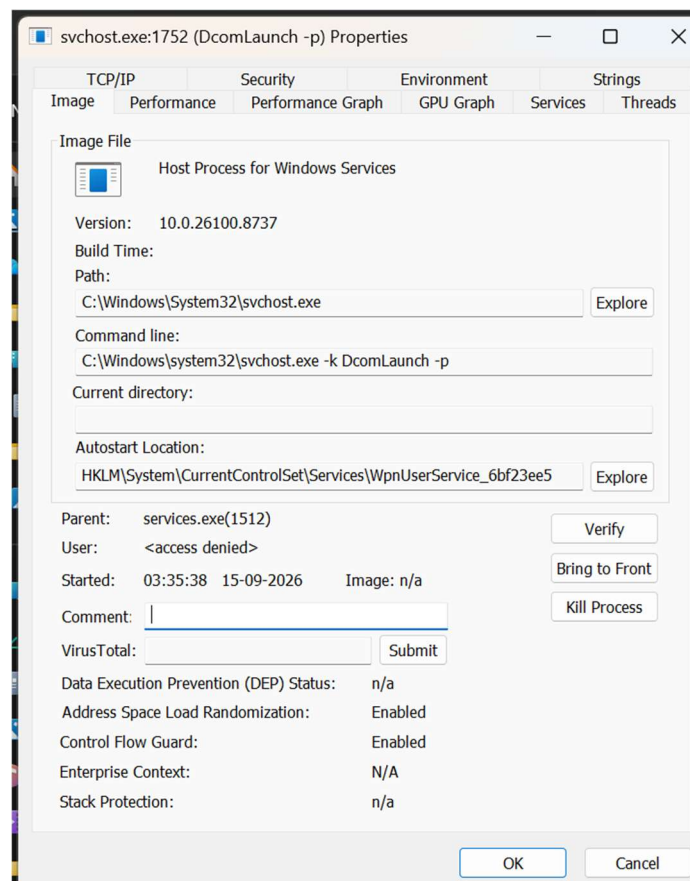
Step 4 – Inspect Process Properties

The `svchost.exe` process was selected and its Properties window was opened.

The Image tab provided the following information:

- Process name: `svchost.exe`
- PID: `1944`
- Description: Host Process for Windows Services
- Path: `C:\Windows\System32\svchost.exe`
- User: `NT AUTHORITY\SYSTEM` ☐ Company: Microsoft Corporation

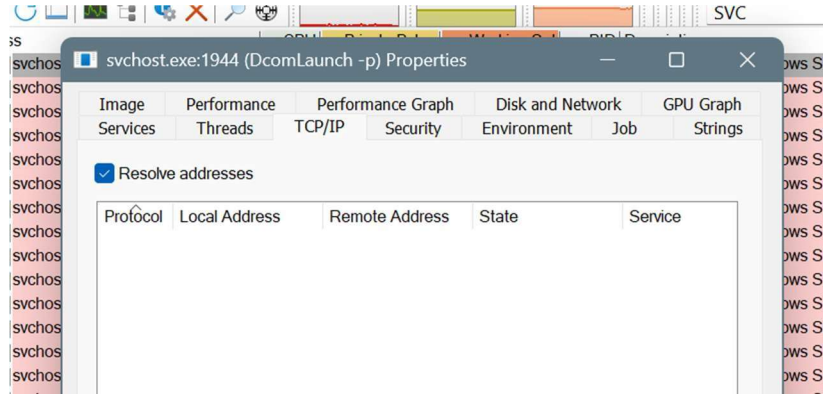
The executable path was inspected to verify that the process was running from the Windows System32 directory.



Step 5 – Inspect TCP/IP Activity

The TCP/IP tab of the 'svchost.exe' Properties window was opened.

The Local Address, Remote Address, Protocol, State and Service information were inspected.



No TCP/IP connections were displayed for the selected process at the time of inspection.

Step 6 – Examine CPU and Memory Usage

The CPU, Private Bytes and Working Set values were examined for the selected 'svchost.exe' process.

The process did not show unusually high CPU usage during the observation.

Step 7 – Online Process Verification

The process name 'svchost.exe' was searched online to understand its purpose.

The search results described 'svchost.exe' as a Windows system process used to host and manage Windows services.

Step 8 – Antivirus Check

Windows Security was opened and the Virus & threat protection section was examined.

The available scan result showed:

- No current threats
- 0 threats found

Scan options

Run a scan from the available options on this page.

No current threats.

Last scan: 23-09-2026 11:49 (quick scan)

0 threats found.

Scan lasted 55 seconds

28945 files scanned.

Quick scan

- Quick scan completed successfully

Result

Process Explorer was successfully used to examine running Windows processes.

The `svchost.exe` process was investigated by checking its:

- Process ID
- CPU usage
- Memory usage
- Description
- Company name
- Executable path
- TCP/IP activity

The investigation showed that the selected `svchost.exe` process was located in the Windows System32 directory and was associated with Microsoft Corporation. No TCP/IP connections were displayed for the selected process during the inspection.

Conclusion

Process Explorer provides detailed information about running Windows processes and their resource usage. It can be used in digital forensics and system investigation to examine process properties, identify unusual resource usage, inspect network activity, and investigate potentially suspicious processes.